

VIGNESH A K

Cellular & Molecular Neuroscience | Neurodegeneration | Cancer Biology | Cell Biology
Bangalore, India | +91 93807 32528 | Vignesh090601@gmail.com | [linkedin.com/in/vigneshak](https://www.linkedin.com/in/vigneshak)

PROFESSIONAL PROFILE

Neuroscience Researcher with 3 years of advanced research experience across premium international institutions, specializing in human iPSC disease modeling, neurodegenerative pathways, and automated high content phenotypic screening workflows. Documented expertise in optimizing multiplexed assays, endolysosomal profiling, and high resolution microscopy. Adept at establishing translationally robust, industry standard in vitro assay architectures and automated image quantification pipelines to accelerate therapeutic compound validation.

CORE COMPETENCIES

- **Cell & Disease Modeling:** Clinical grade hiPSC Culture, Ventral Midbrain Dopaminergic Differentiation, Inducible NGN2 Cortical Neuronal Differentiation, Organotypic Retinal Explants, Mammalian Cell Culture (SOP Management).
- **Assay Development & Screening:** High Content Phenotypic Screening, Multiplexed Immunofluorescence, Live cell Kinetic Assays, Biochemical Assay Optimization, Protocol Scalability & Reproducibility.
- **Microscopy & Imaging Analytics:** Confocal Microscopy, High Resolution Live Cell Fluorescence Imaging (Zeiss ZEN Blue), Automated ImageJ/Fiji Quantitative Data Pipelines, N3 license holder for laser operated microscopes from NEA Singapore
- **Biochemical & Molecular Validation:** Western Blotting (Soluble/Insoluble Fractionation), SDS-PAGE, Size Exclusion Chromatography, Immunocytochemistry, TUNEL Apoptosis Assay, Nanoparticle Tracking Analysis (NTA).
- **Data Analysis & Computational Biology:** GraphPad Prism, RACIPE Framework (Gene-Regulatory Networks), Python, Quantitative rigour & Statistical Profiling.
- **Lab Related:** Trained in Laboratory safety protocol levels 1 and 2 at Hertie Institute. Participated in SOP, protocol writing and proficient in using lab management software such as Labguru for maintaining track of experiments, reporting and lab inventory maintenance.
- **Animal Handling:** LAS Interactive laboratory animal handling certification from LAS interactive GmbH, Germany

RESEARCH EXPERIENCE (3 Years)

Nanyang Technological University, Singapore

Jan 2026 – Sep 2026

Graduate Researcher | Condensate Mediated Regulation of Damaged Lysosome Stabilization and Fate | **Bussi Lab**

- Investigated molecular target mechanisms governing damaged lysosome stabilization and cellular fate under neurodegenerative conditions using high resolution, multichannel live cell imaging platforms.
- Optimized advanced organelle isolation protocols and extracellular vesicle (EV) separation workflows to isolate and profile biomolecular condensates under acute oxidative stress.
- Engineered automated ImageJ/Fiji quantitative analysis macros to extract morphological and spatial metrics of endolysosomal compartments, eliminating user bias and securing data reproducibility for high throughput screening data.

Hertie Institute for Clinical Brain Research, University of Tübingen, Germany

Apr 2025 – Sep 2025

Graduate Researcher | Endolysosomal Dysfunction in an iPSC Model of Corticobasal Syndrome | **Fitzgerald Lab**

- Managed the long term culture of patient derived and control iPSC lines, executing strict QC protocols to ensure pluripotency before scaling differentiation into functional ventral midbrain dopaminergic and inducible NGN2 cortical neurons.
- Validated stage specific neuronal lineage differentiation using MAP2 and DAT biomarker tracking; designed and executed fixed cell multiplexed ICC panels (EEA1, Rab7, LAMP1) to accurately map endocytic traffic alterations.
- Acquired multi dimensional confocal images and processed LysoTracker live cell datasets to statistically model selective neuronal vulnerability based on lysosomal count, volume, and intracellular distribution.

Hertie Institute for Clinical Brain Research, University of Tübingen, Germany

Jan 2025 – Mar 2025

Graduate Researcher | Characterization of Fluid Biomarkers for the Staging of Glial Activation in Mice | **Kaesler Lab**

- Executed multiplexed immunofluorescence staining on thick APPPS1 transgenic mouse brain sections to systematically map neuroinflammatory progression and distinct glial activation stages (ApoE, Iba1, and GFAP coordinates).
- Developed a specialized chemical/spectral autofluorescence quenching protocol that minimized background noise and drastically improved signal-to-noise ratios during high depth confocal image acquisition.
- Fractionated TBS and SDS soluble brain tissue homogenates to perform rigorous western blot assay validation evaluating the sensitivity and cross reactivity of proprietary biomarker antibodies.

Institute of Ophthalmic Research, University of Tübingen, Germany

Sep 2024 – Dec 2024

Graduate Researcher | Organotypic Retinal Explant Cultures for Drug Development | **Durand Lab**

- Maintained ex vivo wild type and rd mutant mouse organotypic retinal explant cultures to systematically evaluate the pharmacokinetic efficacy of neuroprotective drug candidates.

- Quantified photoreceptor degeneration kinetics and survival rates using multiplexed TUNEL apoptosis readouts and cell type specific immunofluorescence profiling (cone arrestin, rhodopsin, Iba1 markers).

Hertie Institute for Clinical Brain Research, University of Tübingen, Germany

Jan 2024 – Jun 2024

*Research Assistant | Biomarker Identification through Extracellular Vesicles in Dementia | **Synofzik Lab***

- Isolated and purified extracellular vesicles (EVs) from patient cerebrospinal fluid (CSF) and iPSC derived neuronal secretomes using size exclusion chromatography columns.
- Conducted high precision particle tracking quality control utilizing Nanoparticle Tracking Analysis (NTA) to analyze EV size distributions, concentration kinetics, and downstream Western blot enrichment patterns.

JSS Science and Technology University & KSOU Mysore, India

Jan 2023 – Jul 2023

*Research Assistant | Synthesis of Copper Chloride Nanoparticles from Lablab purpureus for Phenol Detoxification | **Baker Lab***

- Synthesized copper chloride nanoparticles via green chemistry routes using Lablab purpureus and optimized plant derived peroxidase extraction matrices for industrial scale phenol degradation.

Indian Institute of Science (IISc), Bangalore, India

Jan 2022 – Dec 2022

*Research Assistant | **ELF3** as a Putative MET Inducing Factor | Cancer Systems Biology Laboratory | **Jolly Lab***

- Modeled gene regulatory network dynamics using the RACIPE computational framework to investigate the role of ELF3 in driving epithelial mesenchymal plasticity and phenotypic state transitions.

EDUCATION

M.S. in Cellular and Molecular Neuroscience | Eberhard Karls Universität Tübingen, Germany

Oct 2023 – Sep 2025

Grade: 1.89/1.0 on the German scale (where 1.0 is highest)

B.Engg. in Biotechnology | JSS Science and Technology University, Mysore, India

Aug 2019 – Jul 2023

Grade: 9.9/10.0 CGPA | Honors: Chancellor's Gold Medalist & 1st Rank for University

SELECTED PUBLICATIONS

Subbalakshmi AR, Sahoo S, Manjunatha P, et al., Kasiviswanathan VA, et al. "The ELF3 transcription factor is associated with an epithelial phenotype and represses epithelial-mesenchymal transition." [Journal of Biological Engineering \(2023\)](#).

British Neuroscience Association (2022): "Challenges and Future Prospects of Drug Delivery Across the Blood-Brain Barrier."

College Science Seminars: Industrial Safety Hazards in Rubber Industry (2022), Plant Tissue Culture Techniques for Capsaicin Extraction - Benefits and Challenges (2021).

AWARDS, CREDENTIALS & MEMBERSHIPS

- Chancellor's Gold Medal for highest CGPA in B.Engg. Biotechnology, JSS Science and Technology University (2024).
- Triune Solutions Inc. Endowment Award for outstanding performance in B.Engg. Biotechnology (2024).
- GATE 2023: All India Rank 123; 99.13 percentile.
- N3 license holder for laser-operated microscopes, National Environment Agency, Singapore.
- Student member: British Neuroscience Association, Federation of European Neuroscience Societies, Society for Mathematical Biology.
- Selected training: LAS Interactive laboratory animal handling (2024), Merck Innovation Lab symposia on cell-based functional assays, 2D electrophoresis and biosafety (2022).